

Data visualization

- Data refers to raw facts, figures, or measurements collected about events, objects, or phenomena. It forms the foundation for gaining insights and making informed decisions. Data can be qualitative, such as names or categories, or quantitative, like numerical values or measurements. However, raw data in its initial form is often vast, unstructured, and challenging to interpret.
- Examples of Data:
 - Names of students in a class.
 - Heights of individuals in centimeters.
 - Monthly sales of a product.
 - Customer satisfaction ratings.
- Tabulating and visualizing data are crucial processes that simplify and organize complex datasets, making them easier to understand and analyze.
- Tabulation systematically arranges data into rows and columns, highlighting key features and relationships, while visualization uses charts, graphs, and diagrams to provide a quick, engaging overview of the information.
- Importance of Data Visualization: Data visualization involves using graphical methods to represent data. It plays a critical role in data analysis by enabling:
 - **Clarity:** Simplifies complex datasets, making them easier to understand.
 - **Trend identification:** Highlights patterns and trends that may not be immediately apparent in raw data.
 - **Quick decision-making:** Facilitates faster insights, crucial for decision-making processes.
 - **Engagement:** Visual representation is more engaging and intuitive than raw data or tables.

Organizing data

The process of organizing data related to a quantitative phenomenon typically involves the following stages:

- **Raw data:** A collection of individual observations in their original, unorganized form. For example, a list of exam scores such as 45, 67, 89, 56, etc.
- **Organized (Arrayed) data:** Sorting raw data into ascending or descending order to make patterns more apparent. For instance, arranging the scores as 45, 56, 67, 89.
- **Discrete (Ungrouped) frequency distribution:** Representing data by showing how often each individual value occurs. For example, a table displaying the frequency of each exam score.
- **Grouped frequency distribution:** Combining data values into intervals or ranges (e.g., 40–49, 50–59) and showing the frequency of observations within each range. This method is useful for summarizing large datasets.
- **Continuous frequency distribution:** Similar to grouped frequency distribution but used for continuous data, where the intervals have no gaps (e.g., 40.5–49.5, 50.5–59.5). This is particularly relevant for measurements like height or weight.

Frequency distribution table

- A **frequency distribution** is a tabular arrangement of data that pairs the values of a variable with their corresponding frequencies, i.e., the number of times each value occurs.
- Example: The following table shows the ages of 50 individuals in a ward:

36	30	39	44	40	49	51	41	43	35
30	38	53	32	35	38	32	45	44	41
50	45	42	31	36	50	34	33	32	50
46	39	37	33	34	34	35	48	50	47
35	50	48	45	47	47	43	49	52	35

- Lets express it in the form of a discrete or ungrouped frequency distribution:

Age	Tally Bar	Frequency	Age	Tally Bar	Frequency
30		2	42		2
31		1	43		2
32		3	44		2
33		2	45		3
34		3	46		1
35		5	47		3
36		2	48		2
37		1	49		2
38		2	50		5
39		2	51		1
40		1	52		1
41		1	53		1

Grouped Frequency Distribution

- An ungrouped frequency distribution is particularly useful in two scenarios:
 - When the values of a variable are frequently repeated, making it easier to identify patterns without further grouping.
 - When the variable involves only a small number of distinct values, allowing for direct and clear representation without significant loss of simplicity.
- However, when the values are scattered or numerous, ungrouped frequency distribution may not effectively condense the data. In such cases, the first meaningful step in summarizing the data is dividing it into classes (or class intervals).
- This involves grouping the entire range of variable values into appropriate intervals and recording the number of observations in each group. Grouping data into classes not only simplifies the dataset but also makes the data more interpretable and ready for analysis.
- The following table represents the grouped frequency distribution of the data:

Marks range	Number of students
30-34	11
35-39	12
40-44	8
45-49	11
50-54	8

Grouped frequency distribution

- A grouped frequency distribution can be converted into a **continuous grouped frequency distribution** by ensuring that the intervals are continuous, with no gaps between the upper limit of one class and the lower limit of the next class. This involves adjusting the class boundaries slightly.
 - Identify Class Intervals: Start with the class intervals from the grouped frequency distribution. For example: 30-34, 35-39, 40-44, 45-49, 50-54.
 - Adjust the boundaries of each class to make them continuous. This is done by subtracting a small value from the lower boundary and adding the same value to the upper boundary.
 - For 30-34: new lower boundary = $30 - 0.5 = 29.5$, new upper boundary = $34 + 0.5 = 34.5$. Repeat this for all intervals.
 - Create continuous classes: The adjusted intervals will now be: 29.5-34.5, 34.5-39.5, 39.5-44.5, 44.5-49.5, 49.5-54.5.

Marks range	Number of students
29.5–34.5	11
34.5–39.5	12
39.5–44.5	8
44.5–49.5	11
49.5–54.5	8

- The ideal number of classes in a frequency distribution is essential to ensure the data is represented in a balanced manner. Too few classes can oversimplify the data and obscure important details, while too many classes can make the data difficult to interpret.
- To determine the optimal number of classes, we will use the following formulas:

- $k = 1 + 3.322 \log_{10} N$
- $k = 1 + \log_2 N$

Here, k is the approximate number of classes and N is the total number of observations. **Given Data:**

- Total number of observations, $n = 50$ (from the marks data in the previous example).

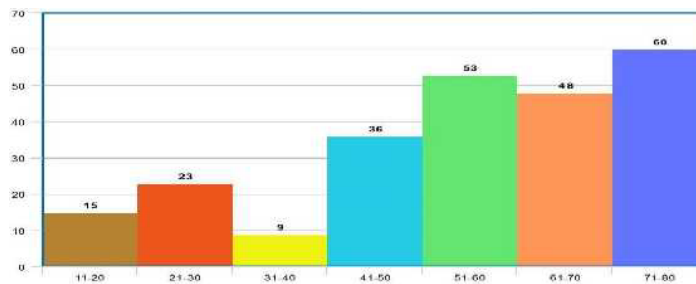
Using formula: $k = 1 + \log_2 50$, $k = 1 + 5.65 = 6.65$. Round k to the nearest whole number: $k \approx 7$. Researchers may adjust the number of classes based on specific dataset characteristics and the objectives of their analysis.

Histogram

- A histogram is a graphical representation of a grouped frequency distribution with continuous classes. It is an area diagram and can be defined as a set of rectangles with bases along with the intervals between class boundaries and with areas proportional to frequencies in the corresponding classes.
- Example: The following table represents the variable and its frequency distribution:

Variable range	Frequency
10–20	15
21–30	23
31–40	9
41–50	36
51–60	53
61–70	48
71–80	60

- Histogram for the above frequency distribution will be like:

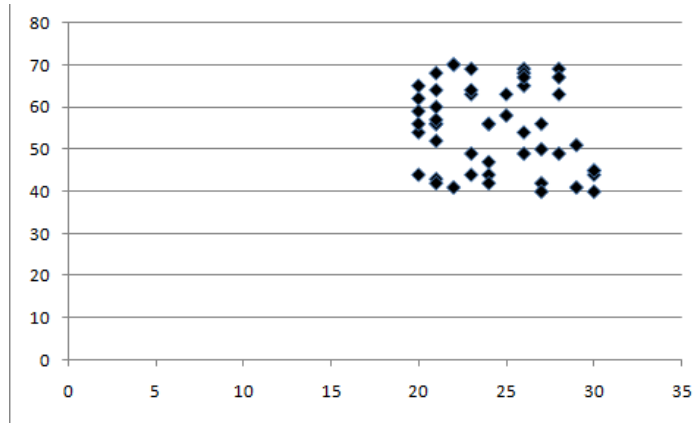


Ogive

- An ogive is a graphical representation of the cumulative frequency distribution of a dataset. It is used to determine the number of observations below a particular value in the dataset and is particularly helpful in understanding the distribution of data. An ogive is a smooth, non-decreasing curve that progresses as the cumulative frequencies increase.

Scattar plot

- Scatter plots are the graphs that present the relationship between two variables in a data-set. It represents data points on a two-dimensional plane or on a Cartesian system. The independent variable or attribute is plotted on the X-axis, while the dependent variable is plotted on the Y-axis. These plots are often called scatter graphs or scatter diagrams.
- Here, a scattar plot is depicted where X-axis shows age and Y-axis shows weights of 50 students.



Cross-Section data

- Cross-section data refers to data collected at a single point in time across multiple subjects or entities. It provides a snapshot of different entities or variables at a particular moment.
- Examples: Income levels of households in a city in 2025, GDP of various countries for the year 2023, test scores of students in a particular class on a single exam day.

Time series data

- Time series data refers to data collected over multiple time periods for a single subject or entity. It captures how a variable changes over time.
- Monthly sales of a product from January to December 2024, temperature readings recorded daily over a year, stock prices of a company observed over a week.

Practice questions

1. A fitness center has conducted a survey to record the number of push-ups completed by 50 participants in a single session. The following data represents the recorded push-ups for each participant: 45, 50, 38, 52, 47, 60, 43, 55, 49, 41, 62, 51, 48, 58, 40, 44, 46, 54, 53, 57, 39, 42, 61, 59, 56, 45, 50, 37, 63, 64, 41, 47, 52, 48, 46, 58, 44, 60, 49, 53, 55, 43, 62, 57, 45, 50, 40, 48, 54, 56. Create an ungrouped frequency distribution table. Organize the data into a grouped frequency distribution table. Use the following intervals for the class ranges: 37–40, 41–44, 45–48, 49–52, 53–56, 57–60, 61–64. Draw a histogram to represent the grouped frequency distribution.
2. A teacher conducted a math exam for 40 students in the class. The teacher wants to analyze the overall performance of the students to identify trends and areas for improvement. The following marks (out of 100) were obtained by the students: 48, 72, 65, 89, 54, 77, 61, 92, 68, 74, 59, 81, 66, 90, 55, 73, 50, 85, 78, 69, 64, 70, 58, 88, 82, 60, 67, 76, 71, 53, 62, 84, 57, 75, 49, 86, 80, 63, 56, 79. Group the marks into class intervals of width 10 and draw histogram and ogive.
3. A sports academy conducted a survey to examine the relationship between the age and height of 20 participants. The following data represents the observations:

Age (Years)	Height (cm)	Age (Years)	Height (cm)
12	140	22	174
13	145	23	175
14	150	24	176
15	152	25	178
16	158	26	179
17	160	27	180
18	165	28	181
19	168	29	182
20	170	30	183
21	172	31	184

Using the given data, draw a scatter plot.

4. A retail store wants to analyze the monthly sales trend for one of its popular products over the past year. The data below shows the sales (in units) for each month:

Month	Sales (Units)	Month	Sales (Units)
January	120	July	170
February	130	August	160
March	140	September	150
April	150	October	140
May	160	November	180
June	170	December	190

Using the data provided, plot a time series graph with: Months on the x-axis and sales (in units) on the y-axis.